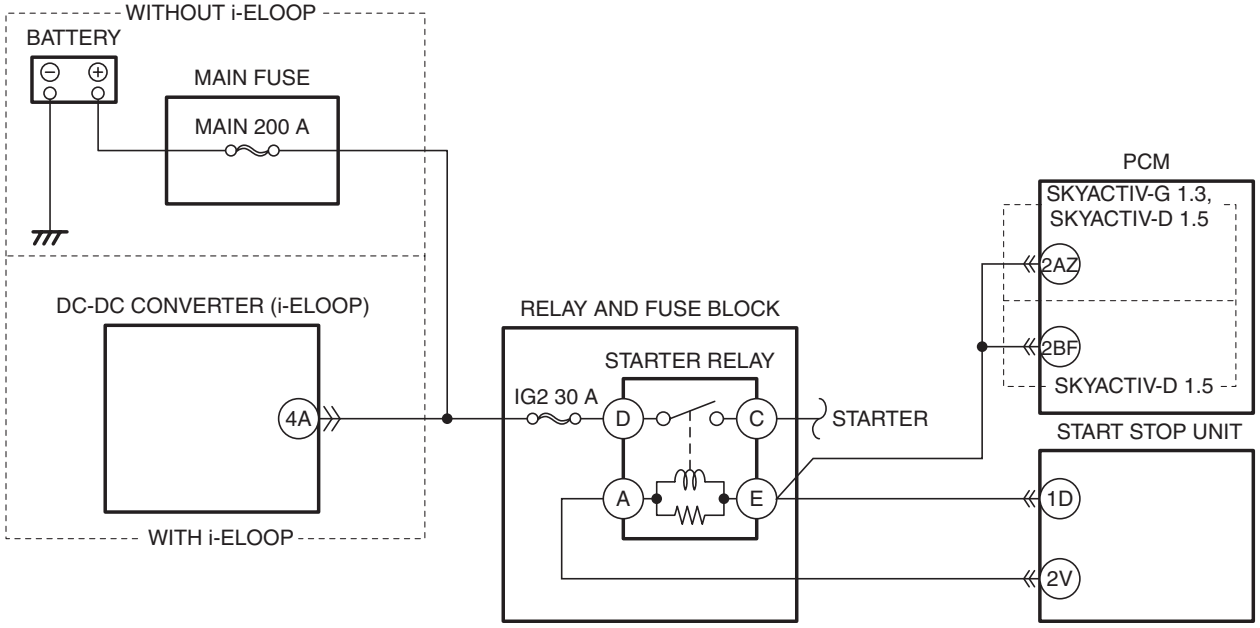
DTC P0615:13 [START STOP UNIT]

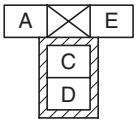
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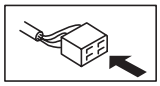
Description	Starter relay circuit malfunction
Detection condition	• Start stop unit detects an open circuit in the starter relay circuit for **0.075 s or more** with the ignition switched off.
Fail-safe	Not applicable
Possible cause	**Note** • If the start stop unit connector is disconnected with the negative battery cable connected, this DTC is output. • Starter relay malfunction • Start stop unit connector or terminal malfunction • Open circuit in wiring harness between the following terminals: — Starter relay terminal E—Start stop unit terminal 1D — Starter relay terminal A—Start stop unit terminal 2V • Start stop unit malfunction

Description	Starter relay circuit malfunction



STARTER RELAY
(RELAY AND FUSE BLOCK)

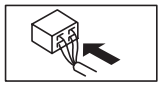




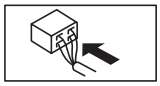
START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR

1AE	1AC	1AA	1Y	1W	1U	1S	1Q	1O	1M	1K	1I	1G	1E	1C	1A
1AF	1AD	1AB	1Z	1X	1V	1T	1R	1P	1N	1L	1J	1H	1F	1D	1B

2W	2U	2S	2Q	2O	2M	2K	2I	2G	2E	2C	2A
2X	2V	2T	2R	2P	2N	2L	2J	2H	2F	2D	2B



DC-DC CONVERTER (i-ELOOP)
WIRING HARNESS-SIDE CONNECTOR

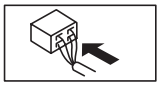


PCM
WIRING HARNESS-SIDE CONNECTOR

2BE	2AZ	2AU	2AP	2AK
2BF	2BA	2AV	2AQ	2AL
2BG	2BB	2AW	2AR	2AM
2BH	2BC	2AX	2AS	2AN
	2BD	2AY	2AT	2AO

2AE	2AA	2W	2S	2O	2K	2G	2C
2AF	2AB	2X	2T	2P	2L	2H	2D

2AI	2AG	2AC	2Y	2U	2Q	2M	2I	2E	2A
2AJ	2AH	2AD	2Z	2V	2R	2N	2J	2F	2B



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT STARTER RELAY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the starter relay. (See RELAY LOCATION.) • Inspect the starter relay. (See RELAY INSPECTION.) • Is the starter relay normal?	Yes	Go to the next step.
		No	Replace the starter relay, then go to Step 4. (See RELAY LOCATION.)
2	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 4.
3	**INSPECT STARTER RELAY CIRCUIT FOR OPEN CIRCUIT** • Verify that the starter relay is removed. • Verify that the start stop unit connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Starter relay terminal E—Start stop unit terminal 1D — Starter relay terminal A—Start stop unit terminal 2V • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Starter relay terminal E—Start stop unit terminal 1D • Starter relay terminal A—Start stop unit terminal 2V **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
4	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition off and wait for **1 s or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
5	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.